

A novel approach to characterizing the visual dimensions of mental imagery vividness using Gaussian process regression

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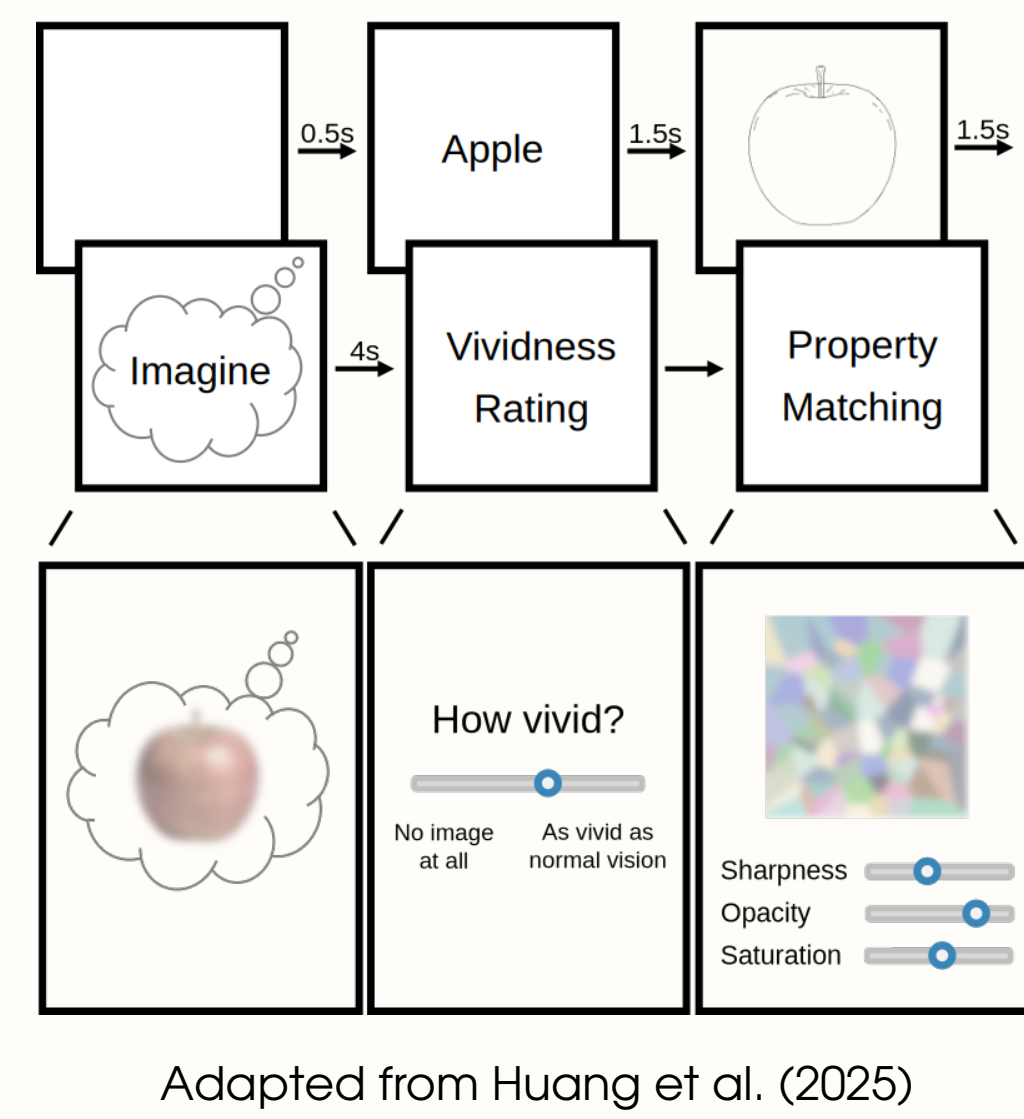
(1) How to characterise vividness?

Huang et al. (2025) used linear regression to reveal evidence of common structure in 'vividness' & large individual differences.

Current aims

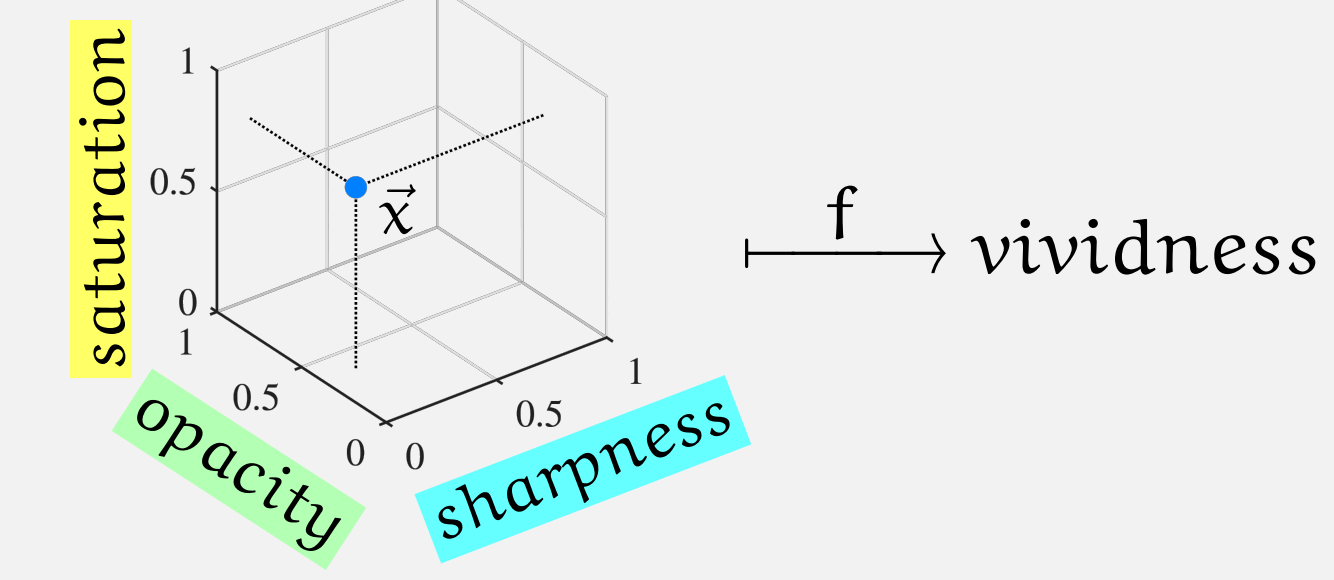
- (1) Move beyond the parametric assumptions of linear models
- (2) Characterise individual differences in a principled manner

(2) Data and problem formulation



Vividness ratings (1 – 5 continuous scale)
Perceptual features (0 – 1 continuous scale): sharpness, opacity, saturation

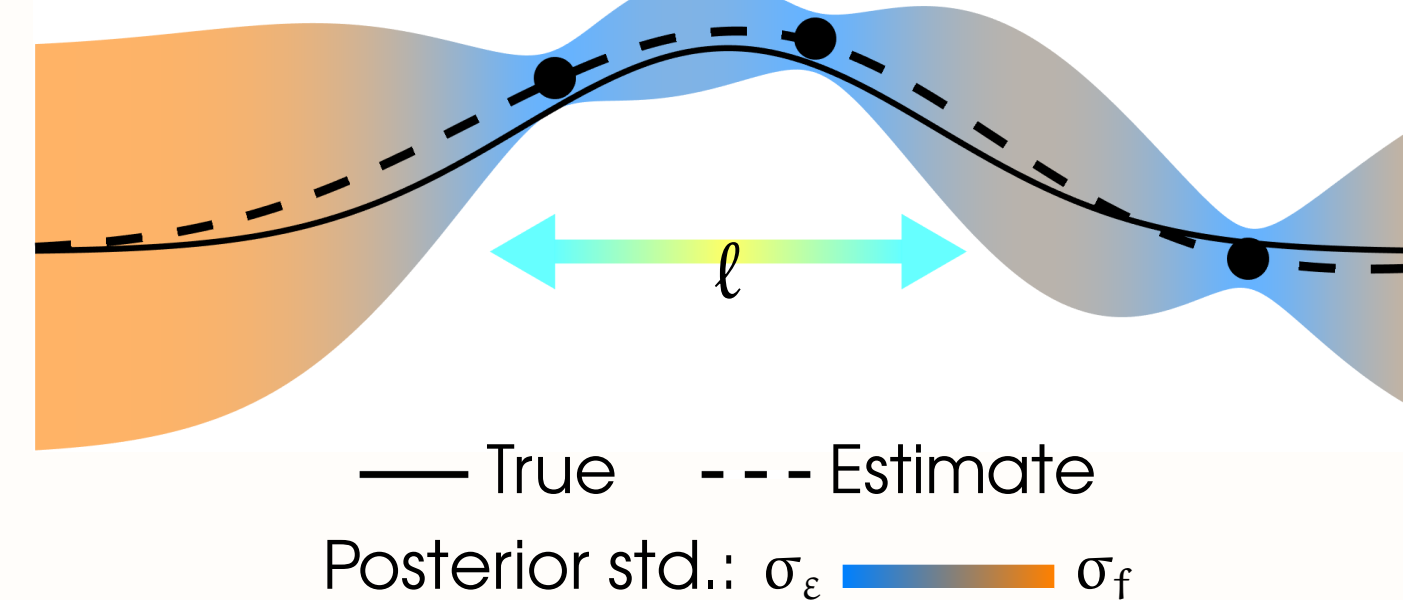
What is the function $f(\cdot)$ that maps these features onto vividness?



(3) What are Gaussian processes?

Setup: N observations y_i of $f(\cdot)$ at input locations $\vec{x}_i$ such that $y_i = f(\vec{x}_i) + \epsilon_i$

Core idea: Finite sets of evaluations of $f(\vec{x})$ are jointly Gaussian with the covariance structure determined by inputs $\vec{x}$ (Rasmussen and Williams 2005)



$$\text{Cov}[y_i, y_j] = k(\vec{x}_i, \vec{x}_j) + \sigma_\epsilon^2 \delta_{ij}$$

$$= \sigma_f^2 \exp \left(-\frac{1}{2} \left(\frac{(\vec{x}_i^{\text{sha}} - \vec{x}_j^{\text{sha}})^2}{\ell_{\text{sharpness}}^2} + \frac{(\vec{x}_i^{\text{opa}} - \vec{x}_j^{\text{opa}})^2}{\ell_{\text{opacity}}^2} + \frac{(\vec{x}_i^{\text{sat}} - \vec{x}_j^{\text{sat}})^2}{\ell_{\text{saturation}}^2} \right) \right) + \sigma_\epsilon^2 \delta_{ij}$$

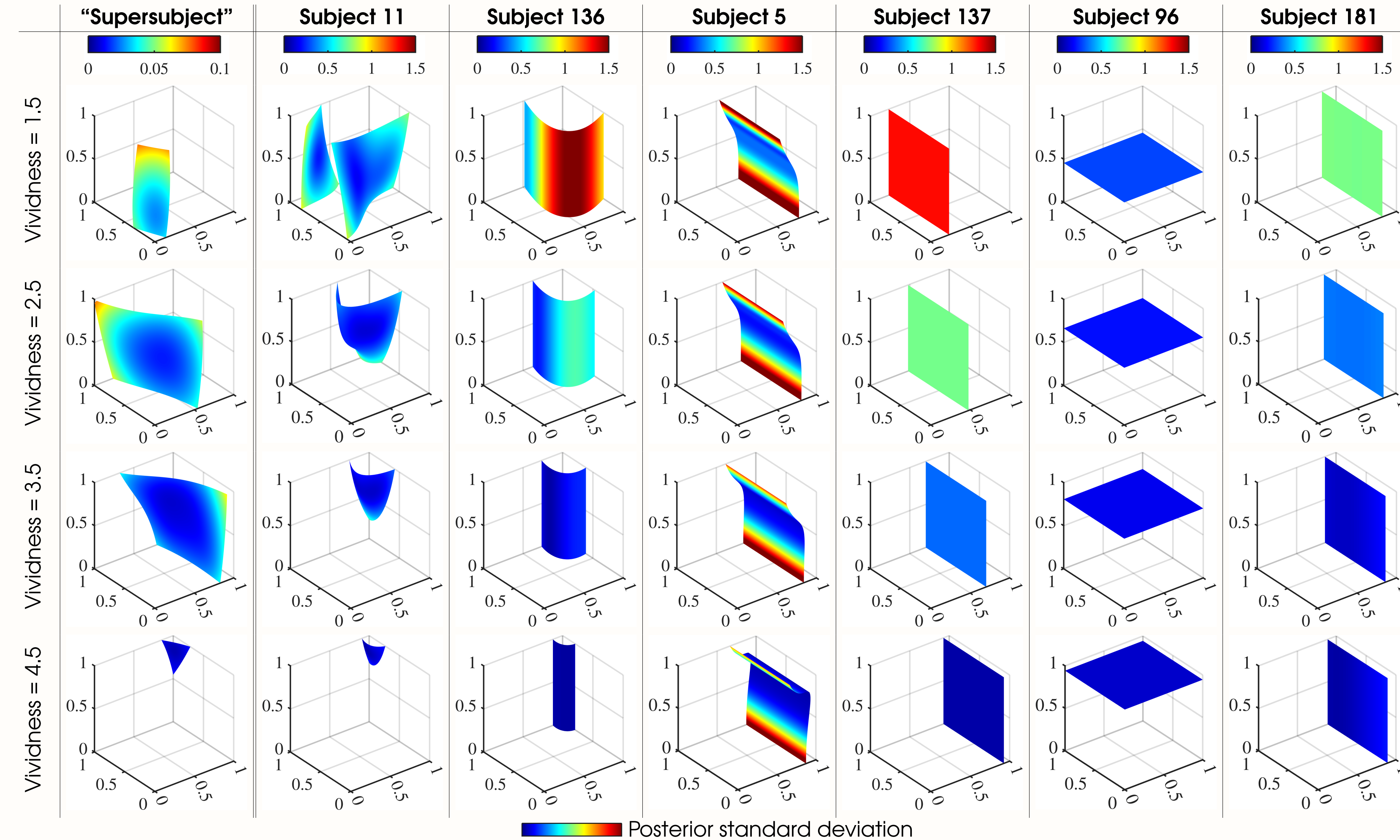
where δ_{ij} is the impulse function that takes the value 1 only when $i = j$ and 0 otherwise.

Model fits are obtained using MATLAB Statistics and Machine Learning Toolbox (2026).

Hyperparameters:

- **Kernel function:** Covariance between noiseless function evaluations
- **Noise standard deviation:** Uncertainty on measurements $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$
- **Signal standard deviation:** Uncertainty on unobserved locations
- **Length scales:** Smoothness over variables

(4) Gaussian processes reveal nonlinearities and clusters of individuals

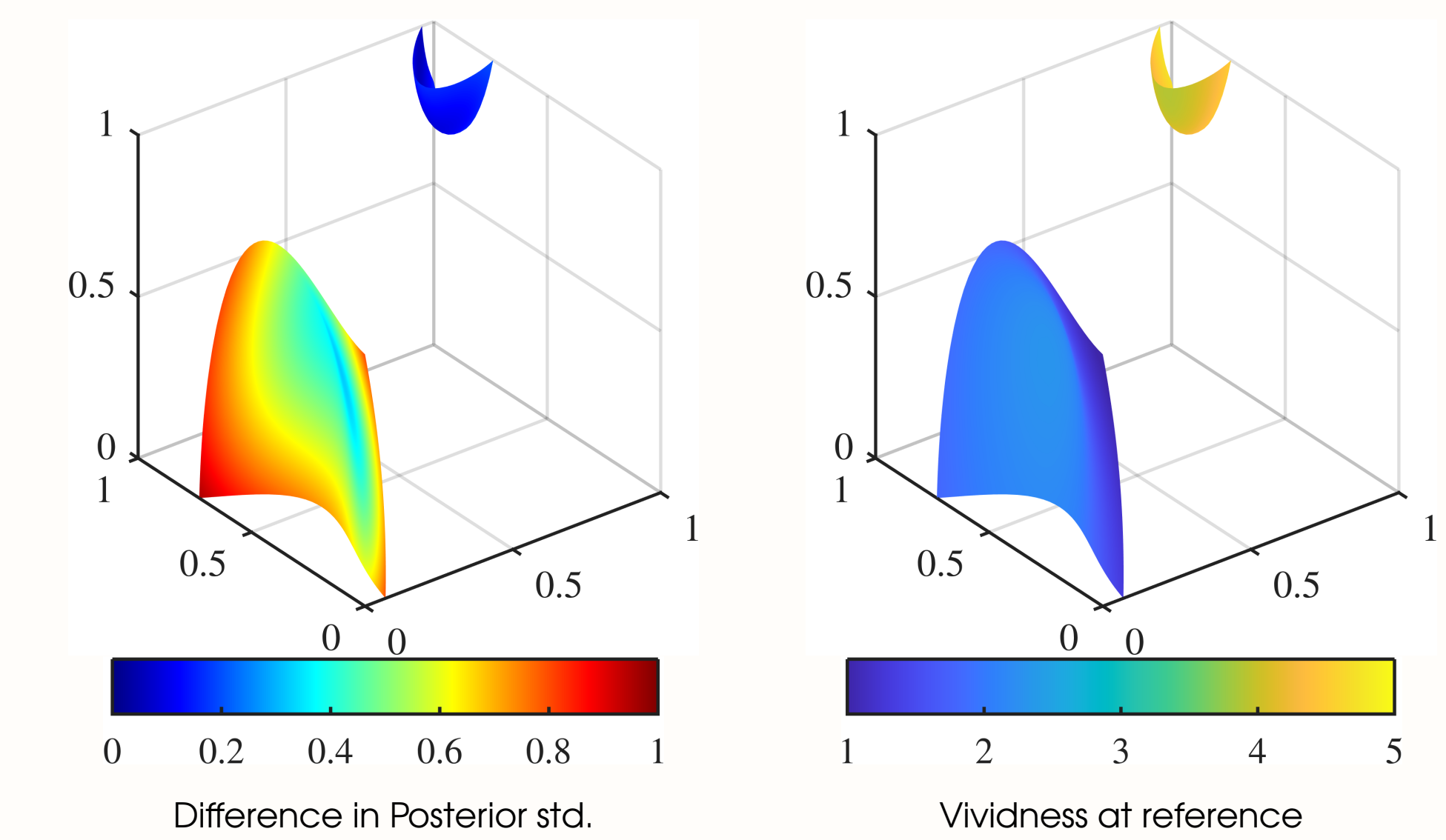


Equal vividness manifolds

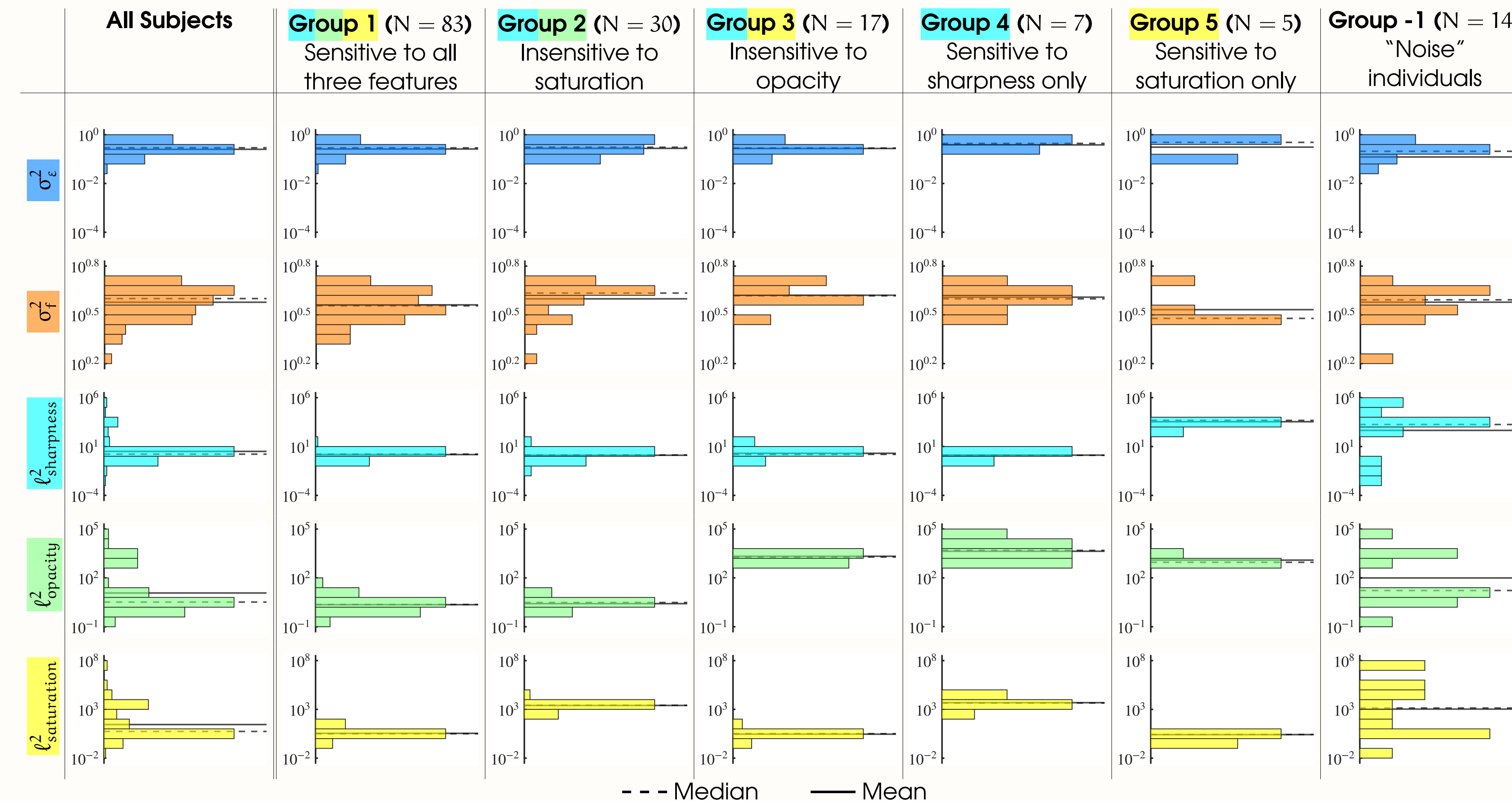
where different features correspond to the same vividness rating

Difference manifolds

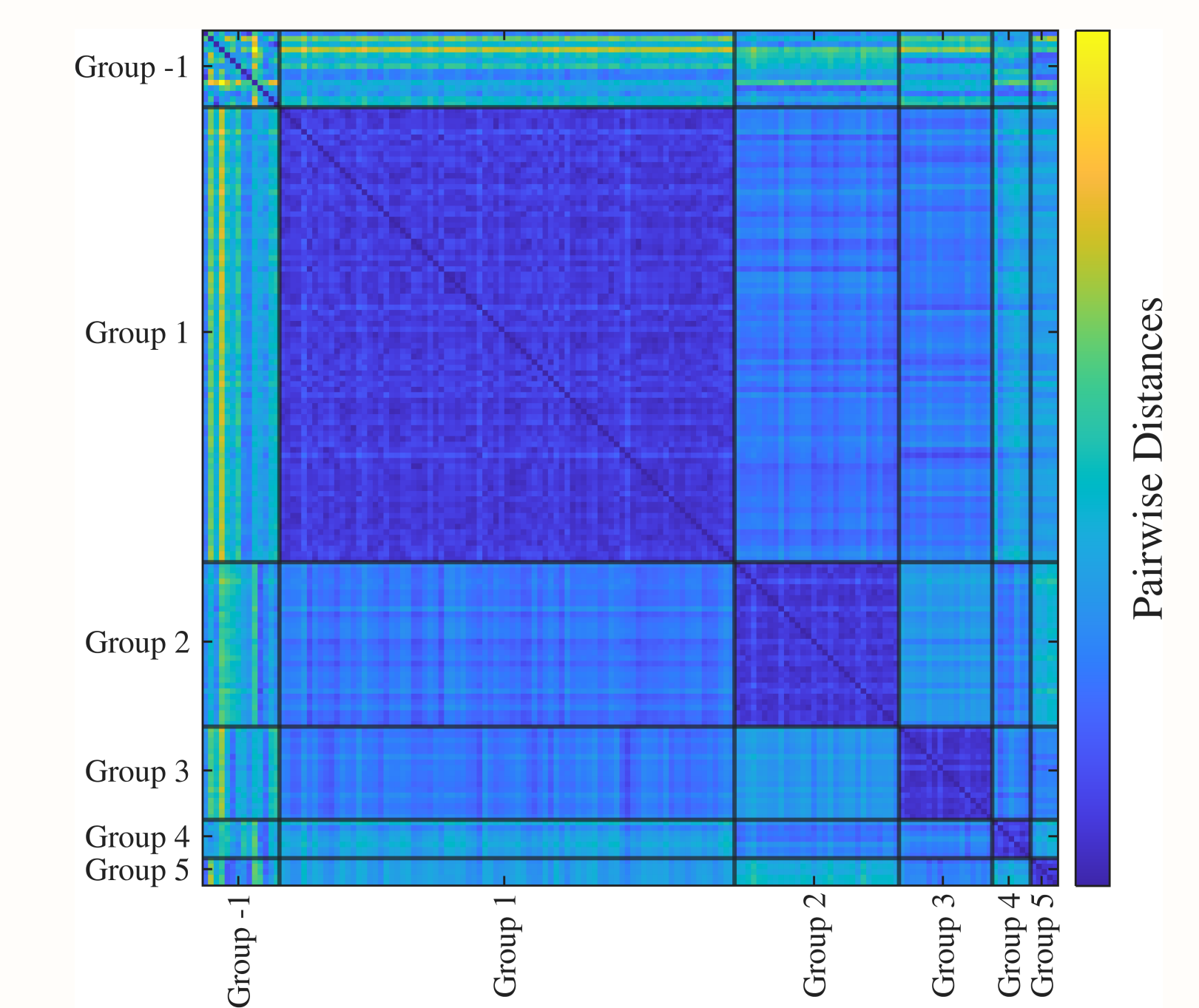
where the difference Δ between two profiles is constant



Non-planar manifolds imply that nonlinear relationships can be captured using Gaussian processes!



Hyperparameter-based clustering



Clustered using DBSCAN (Ester et al. 1996) and Satopaa et al. (2011) for knee detection.

Groups differ mostly in the way they are sensitive to the three perceptual features!

(5) Take-home messages

Vividness ratings...

- (1) can show nonlinear dependence on perceptual features, as seen through non-planar equal vividness manifolds over the feature space!
- (2) show individual differences in their dependence on these features, agreeing with Huang et al. (2025), ...yet there are trends in these differences!

Why use Gaussian processes?

- Non-parametric regression method not restricted by functional form
- Inherently generative model
- Model confidence/prediction intervals come naturally
- Not a black box model: hyperparameters are interpretable

Gaussian processes can help answer...

- How do individuals differ from the general population? From each other?
- How to choose input values to elicit a specific response for any multivariate nonlinear behavioral output, without parametric assumptions?
- ... and in a way that agrees with the whole population?

References

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